Esercizio 5.1. Calcolare il determinante delle matrici seguenti riducendole a forma triangolare mediante mosse di Gauss.

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 1 \\ 3 & 1 & 2 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 2 & 0 & 2 & 1 \\ 1 & 3 & 2 & 1 \\ 0 & 2 & 3 & 1 \\ 3 & 1 & 2 & 1 \end{bmatrix}, \qquad \mathbf{C} = \begin{bmatrix} 4 & 0 & 0 & 1 & 1 \\ 1 & 1 & 2 & 0 & 1 \\ 0 & 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{bmatrix} \tag{5.1}$$

1 Ridurre per righe:

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 1 \\ 3 & 1 & 2 \end{pmatrix} \mapsto \begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & -5 \\ 0 & -5 & -7 \end{pmatrix} \mapsto \begin{pmatrix} 1 & 2 & 3 \\ 0 & -5 & -7 \\ 0 & 0 & -5 \end{pmatrix} = U$$

$$\det(A) = (-1)^s \prod_{i=1}^{3} u_{ii} = (-1)^1 \, 25 = -25.$$

Laplace

$$\begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 1 \\ 3 & 1 & 2 \end{vmatrix} = 3C_{31} + 1C_{32} + 2C_{33}$$

$$= 3 \cdot (-1)^{3+1} \begin{vmatrix} 2 & 3 \\ 4 & 1 \end{vmatrix} + 1 \cdot (-1)^{3+2} \begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix} + 2 \, (-1)^{3+3} \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix}$$

$$= 3 \cdot 1 \cdot (-10) + 1 \cdot (-1)(-5) + 2 \cdot 1 \cdot 0$$

$$= -30 + 5 + 0$$

$$= -25$$

Laplace (riga):

$$\begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 1 \\ 3 & 1 & 2 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 \\ 0 & 0 & -5 \\ 0 & -5 & -7 \end{vmatrix} = 1C_{11} = 1 \cdot (-1)^{1+1} \begin{vmatrix} 0 & -5 \\ -5 & -7 \end{vmatrix} = 1 \cdot 1 \cdot (-25) = -25$$